

CODE ALPHA – TASK 2

Modern Web-Based Calculator – Project Report

1. Introduction

The Smart Calculator Pro is a modern web-based application developed using HTML, CSS, and JavaScript. It performs basic arithmetic operations while providing an enhanced user experience through a clean interface, responsive design, and additional smart features such as keyboard support, theme switching, and calculation history.

This project demonstrates fundamental frontend development skills along with UI/UX improvements to make the application more interactive and user-friendly.

2. Objective

The main objectives of this project are:

- To design a functional calculator using web technologies
 - To implement arithmetic operations (+, −, ×, ÷)
 - To create a clean and responsive user interface
 - To handle user input efficiently
 - To enhance usability with additional features like keyboard input and history tracking
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3. Tools and Technologies Used

- **HTML** – Structure of the calculator
- **CSS** – Styling, layout, and responsiveness
- **JavaScript** – Logic, calculations, and interactivity

4. Features of the Project

Basic Functional Features

- Performs addition, subtraction, multiplication, and division
- Display screen for input and output
- Clear (C) and delete (⌫) functionality
- Handles invalid inputs with error display

Advanced Features

- Keyboard support for faster input
- Light/Dark theme toggle for better user experience
- Calculation history panel
- Persistent storage using localStorage
- Responsive design for different screen sizes

5. Working of the Project

The calculator works by capturing user input through button clicks or keyboard events. The entered values are displayed on the screen. When the user presses the equals (=) button, the expression is evaluated using JavaScript logic, and the result is displayed.

The application also stores each calculation in a history panel, allowing users to review previous operations. The theme toggle feature allows switching between light and dark modes dynamically.

6. User Interface Design

The interface is designed using modern UI principles:

- Glassmorphism effect for a stylish look

- Gradient background for visual appeal
- Clearly arranged buttons in grid format
- Highlighted operator and equals buttons
- Smooth hover and click animations

This ensures the calculator is both functional and visually appealing.

7. Output

The final output is a fully functional calculator that:

- Accepts user input via buttons and keyboard
 - Performs real-time calculations
 - Displays results instantly
 - Maintains a history of calculations
 - Provides a smooth and responsive user experience
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8. Conclusion

The Smart Calculator Pro successfully fulfills the objective of building a functional and interactive calculator using web technologies. In addition to basic operations, the project includes modern UI design and enhanced features such as history tracking and theme switching, making it more user-friendly and suitable for real-world usage.